

Supplementary Information:

Metal bottlenecks along energy transitions call for
technology flexibility and sobriety

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1 Supplementary Results

1.1 Comparing cumulated demand to reserves

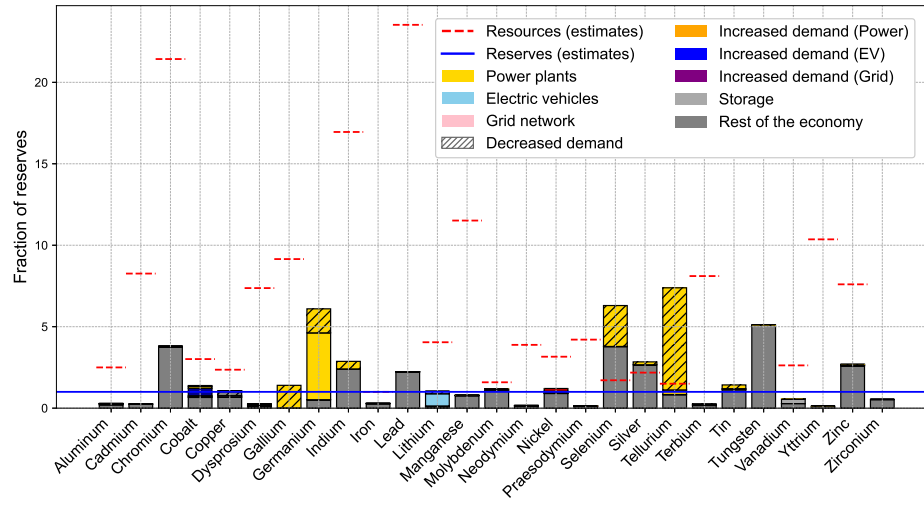


Fig. 1 Global cumulated metal demand from 2020 to 2050 per metal and economic sector aggregates, compared to estimated reserves and resources [1, 2]. The reserves are normalized at 1 (=100%) and normalized metal demand and resources are calculated as a fraction of reserves. For power plants, EVs and the electric grid, the pastel colors represent the initial IAM demands, while the hatched parts represent a decrease and the darker colors an increase in metal demand because of the optimization of technological market shares. In grey are the demands that are identical between both scenarios.

1.2 Resource depletion for the 110 scenarios

Table 1 Percentage of the 110 models constrained by resources in the initial and optimized scenario. An overshoot is considered if the cumulated demand is above resources. A societal effort is considered to be necessary if the demand from the transition sectors are above more than 10% of the resources.

Metal	Overshoot		Societal effort	
	Initial	Optimized	Initial	Optimized
Aluminum	0	0	0	0
Cadmium	0	0	0	0
Chromium	0	0	0	0
Cobalt	0	0	0	0
Copper	0	0	0	0
Dysprosium	0	0	0	0
Gallium	0	0	0	0
Indium	0	0	0	0
Iron	0	0	0	0
Lead	0	0	0	0
Lithium	0	0	0	0
Manganese	0	0	0	0
Molybdenum	0	0	0	0
Neodymium	0	0	0	0
Nickel	0	0	0	0
Praesodymium	0	0	0	0
Selenium	100%	100%	5%	100%
Silver	100%	99%	81%	97%
Tellurium	100%	1%	0	0
Terbium	0	0	0	0
Vanadium	0	0	0	0
Yttrium	0	0	0	0
Zinc	0	0	0	0

1.3 Annual metal demand overshoot for the 110 scenarios

Table 2 Percentage of models constrained by refined capacity in the initial and optimized scenarios. A constraint is considered if annual metal demand is above refined capacity, and that the demand from the transition sectors is above 10% of the refined capacity.

Metal	Initial scenario			Optimized scenario		
	2030	2040	2050	2030	2040	2050
Aluminum	6%	0	0	5%	0	0
Boron	0	0	0	0	0	0
Cadmium	9%	28%	28%	0	0	0
Chromium	0	0	0	0	0	0
Cobalt	0	5%	0	0	0	0
Copper	100%	100%	100%	5%	90%	100%
Dysprosium	64%	63%	37%	2%	5%	4%
Gallium	41%	45%	45%	0	0	2%
Germanium	5%	30%	37%	0	0	0
Hafnium	2%	0	0	0	0	0
Indium	13%	42%	45%	0	0	0
Iron	0	0	0	0	0	0
Lead	0	0	0	0	0	0
Lithium	10%	87%	65%	10%	87%	0
Manganese	0	0	0	0	0	0
Molybdenum	28%	15%	10%	4%	25%	22%
Neodymium	12%	15%	0	0	0	0
Nickel	0	0	0	0	0	0
Niobium	0	0	2%	0	2%	1%
Praesodymium	1%	0	0	0	0	0
Selenium	2%	32%	45%	0	0	0
Silicon	2%	0	0	0	0	0
Silver	9%	21%	13%	0	0	2%
Tantalum	0	0	0	0	0	0
Tellurium	25%	80%	87%	0	1%	0
Terbium	55%	53%	30%	0	2%	1%
Tin	62%	82%	85%	1%	2%	5%
Tungsten	0	0	0	0	0	0
Vanadium	15%	100%	100%	15%	100%	100%
Yttrium	0	0	0	0	0	0
Zinc	6%	3%	0	2%	2%	0
Zirconium	0	0	0	0	0	0

1.4 Power sub-technology increase for the 110 scenarios

Table 3 Percentage of the 110 models which shows an increase in the installation of the power plant after optimization.

Power plant	2030	2040	2050
c-Si (total)	10%	8%	7%
Silver	6%	5%	2%
Copper	98%	98%	98%
Thin Film	4%	8%	5%
CSP (total)	90%	95%	92%
Ag-GaAs	48%	46%	41%
Al-MgF2	98%	98%	98%
Wind PM (total)	13%	34%	46%
Offshore	62%	88%	91%
Onshore	3%	15%	25%
Wind wo PM (total)	36%	65%	76%
Offshore	71%	95%	96%
Onshore	27%	37%	42%
Nuclear	49%	25%	14%
Hydro	23%	25%	36%
Biomass	65%	94%	93%
Geothermal	65%	74%	80%

1.5 Motor sub-technology increase for the 110 scenarios

Table 4 Percentage of the 110 models which shows an increase in the installation of the motor after optimization.

Motor	2030	2040	2050
Permanent Magnet (total)	16%	0%	0%
Cu	0%	0%	0%
Al	98%	98%	98%
Induction (total)	82%	98%	98%
Cu	74%	89%	96%
Al	98%	98%	98%

1.6 Battery sub-technology increase for the 110 scenarios

Table 5 Percentage of the 110 models which shows an increase in the installation of the battery after optimization.

Battery	2030	2040	2050
Gr-LMO	65%	98%	98%
Gr-NMC	89%	13%	98%
Gr-LMFP	0%	0%	0%
Gr-NCA	38%	13%	98%
Gr-LFP	9%	85%	0%
Na-ion	9%	86%	98%

1.7 Battery market shares

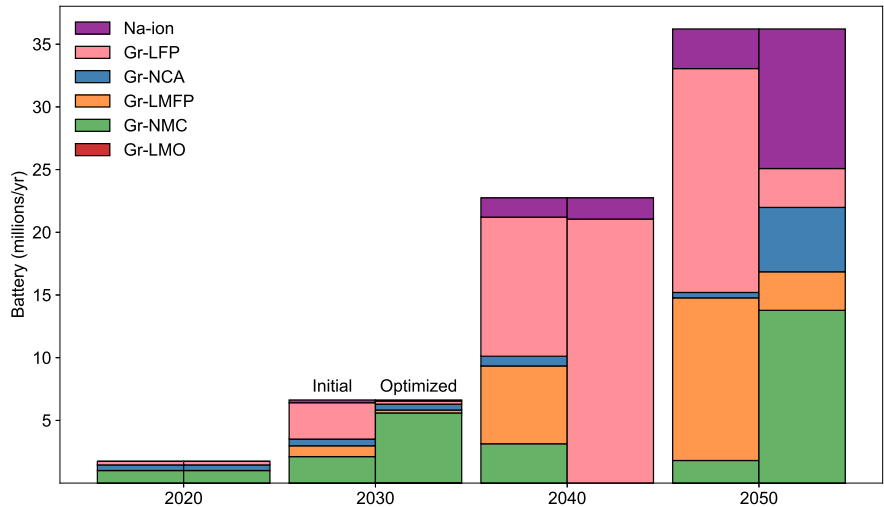


Fig. 2 Comparison of the market share mix of batteries in EVs required to meet the light duty vehicle demand projected by IMAGE for the SSP1-26 scenario, combined with the estimated share of EVs from the Announced Pledges IEA Scenario (APS). On the left, the initial market share mix based on the most likely sub-technological market shares from the literature. On the right, the optimized market share of batteries, designed to avoid exceeding material constraints.

1.8 Power plant market shares for SSP2, SSP4 and SSP5

The optimization of the initial IAM market share of power plants, EVs and grid has been conducted under SSP2, SSP4 and SSP5, by the IAM marker of each narrative, under RCP 2.6. Since the optimized market shares across all SSPs of the electric vehicle (EV) batteries, engines and the electric grid are almost identical to the SSP1 results, they are not included to these supplementary results. For power plant market shares, the general tendencies presented for SSP1 remain similar between SSPs, but with variations in the exact results.

In all scenarios, there is no new installation of thin film and of crystalline silicon (c-Si) with silver conductors after optimization, and a substitution of around 40% of silver by copper conductors. The use of solar is decreased across all SSPs, except in SSP5, which has an important share of concentrated solar power in its optimized mix. Since SSP2 and SSP4 have no concentrated solar power (CSP) installed in the initial scenario, the optimized share of CSP remains limited. The alternative CSP technology with Al and MgF2 is preferred across all SSPs to the Ag-GaAs one. In all SSPs scenarios, the use of wind is greatly increased. The PM turbines are decreased and replaced by gearbox ones in the short term. Though, we notice an increasing share of PM turbines in 2050 in the SSP4. Biomass capacity increases across all scenarios, but more significantly in SSP4 and SSP5. Since nuclear represented an important share of the SSP4 mix, it is greatly decreased after optimization. Though, the final optimized share of nuclear is similar across all SSPs. Hydroelectricity is always decreased, except for a limited expansion in SSP4.

The general tendencies remain similar between SSPs, but SSP4 and SSP5 seem to require more adjustments in terms of installed power capacities in comparison to the initial scenario. This difference can be explained by a more important energy requirement in both scenarios.

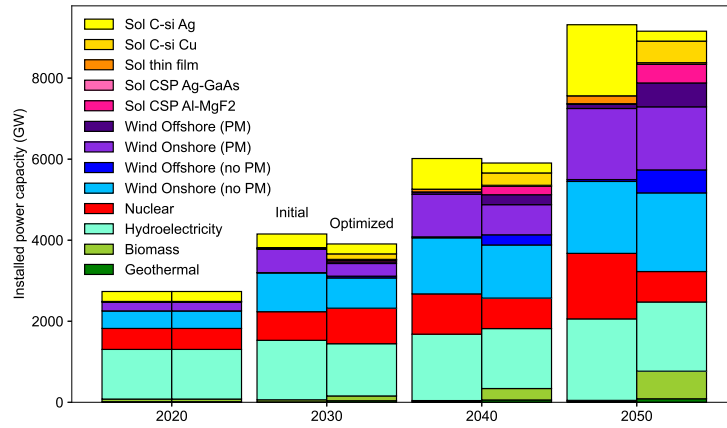


Fig. 3 Comparison of the market share mix of cumulated installed power capacities required to meet the power plant demand projected by MESSAGE-GLOBIOM for the SSP2-26 scenario. On the left, the initial market share mix based on MESSAGE-GLOBIOM's power plant mix estimates and the most likely sub-technological market shares from the literature. On the right, the optimized market shares of power plant sources, designed to avoid exceeding material constraints.

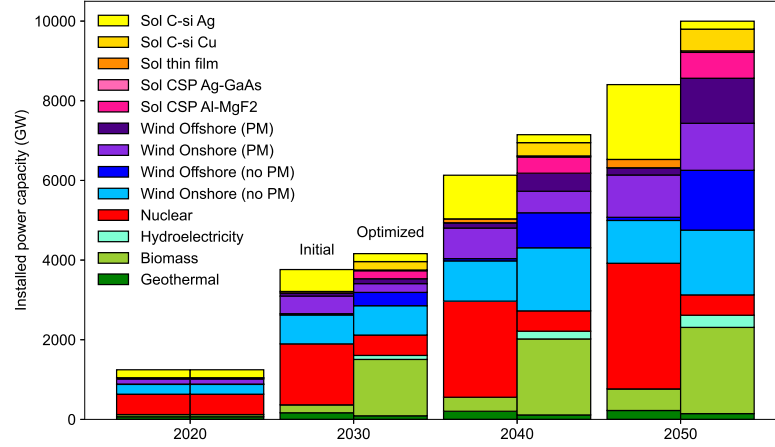


Fig. 4 Comparison of the market share mix of cumulated installed power capacities required to meet the power plant demand projected by GCAM4 for the SSP4-26 scenario. On the left, the initial market share mix based on GCAM4's power plant mix estimates and the most likely sub-technological market shares from the literature. On the right, the optimized market shares of power plant sources, designed to avoid exceeding material constraints.

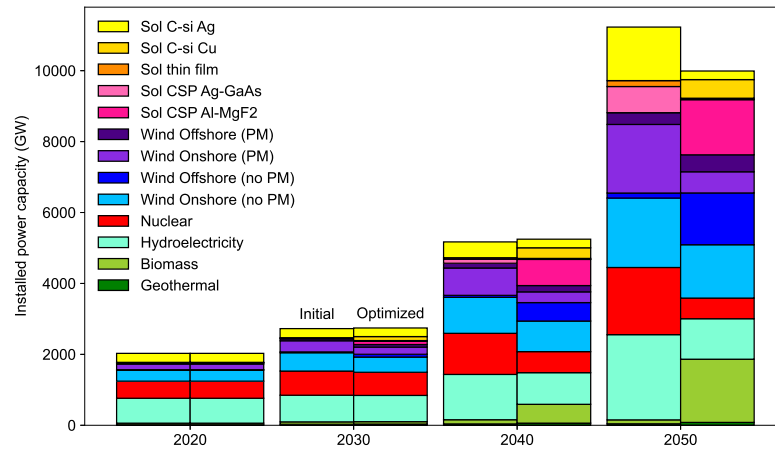


Fig. 5 Comparison of the market share mix of cumulated installed power capacities required to meet the power plant demand projected by REMIND-MAGPIE for the SSP5-26 scenario. On the left, the initial market share mix based on REMIND-MAGPIE's power plant mix estimates and the most likely sub-technological market shares from the literature. On the right, the optimized market shares of power plant sources, designed to avoid exceeding material constraints.

507 1.9 Sensitivity analysis on β

508 The estimation of future refinery capacity for most metals is based on the annual
509 growth parameter β , derived from the historical production growth rate of each metal.
510 Given the uncertainty surrounding this assumption, we performed a sensitivity analysis
511 to assess the influence of β on our results. Refining capacities reported by the IEA
512 are considered robust and therefore kept unchanged (copper, cobalt, nickel, lithium).
513 Consequently, by-product capacities linked to copper estimates also remain unchanged
514 (selenium, tellurium).
515

516 We first fixed the β value at 4.6% for all metals, corresponding to the average
517 β across all metals. With this adjustment, the heatmaps of metal availability risks
518 remained very similar to the main model. In the initial scenario, only tin appeared
519 less constrained under SSP2 and SSP4. After optimization, the same bottlenecks (red
520 markers) were identified, with the exception of dysprosium, which became critical in
521 2050 under SSP5. Final market shares for grid, motor, and battery technologies were
522 nearly identical across all SSP scenarios. Power plant market shares also displayed the
523 same prioritization trends by technology, with only minor differences: a slightly higher
524 share of CSP and geothermal in SSP2, and a reduced share of wind in SSP5. Overall,
525 our main conclusions were unchanged.

526 Next, we increased β by +1%, +2%, and +3% for all metals. Even with a 3%
527 increase, results showed very limited changes. In the initial scenario, bottlenecks on
528 dysprosium and terbium disappeared when β increased by 2%, and fewer metals
529 required societal efforts. However, the set of bottlenecks identified after optimization
530 remained unchanged. Market share trends also persisted, though nuclear and thin-
531 film technologies retained a slightly larger role in the final mix, and biomass shares
532 decreased somewhat due to the greater refining capacity.

533 Finally, we reduced β by -1%, -2%, and -3% to simulate less optimistic mining
534 capacity increase, while imposing a minimum of 0.87%, corresponding to the low-
535 est historical β calculated. With a 1% decrease, the metals appearing as bottlenecks
536 remained similar after optimization. Few changes were observed for the technological
537 market shares, with almost identical results for motor, battery and grid. The final
538 market share of CSP and biomass were slightly more important, with a small decrease
539 for wind. The general tendencies in comparison to the initial scenario remain similar
540 to the main modelisation. However, a 2% decrease in mining capacity revealed more
541 significant impact: neodymium, dysprosium, germanium, and silver emerged as bot-
542 tlenecks in the initial scenario, and optimization failed to resolve capacity issues for
543 dysprosium. In consequence, we observe a more important share of CSP and biomass in
544 the final mix, while wind installations decreased in comparison to the initial scenario.

545 In summary, variations in β had limited influence on our key findings and recom-
546 mendations. Only a reduction of 2% or more in the assumed growth rate produced
547 critical bottlenecks, notably for dysprosium, that could not be resolved through opti-
548 mization. Such constraints would necessitate a stronger deployment of CSP and
549 biomass, and a reduction in wind installation. However, since rare earth mining growth
550 rates of 2–3% per year underpin this outcome, and other studies converge on more
551 optimistic estimates around 5%, this scenario remains unlikely [3, 4].
552

1.10 Metal bottlenecks with societal efforts from the rest of the economy

The heatmap on metal bottlenecks have been conducted with a demand from the rest of the economy assumed proportional to population growth, estimated by IAMs [5], rather than GDP growth. Thus, we made the assumption of the metal demand per capita constant until 2050, simulating sobriety efforts from the rest of the economy.

In the initial scenario, dysprosium, germanium, terbium and tin are not constrained by mining capacity anymore. Almost no metals have more than 90% of the annual demand stemming from the rest of the economy. Results for resources and reserves are quite similar than in the GDP growth scenario. After optimization, copper and vanadium remain constrained by mining capacity even by reducing the demand from the rest of the economy. Only three metals remain constrained by mining capacity with more than 90% of the demand coming from the rest of the economy: niobium, selenium and tellurium. No metals are constrained by resources anymore and results for reserves are quite similar than in the GDP growth scenario.

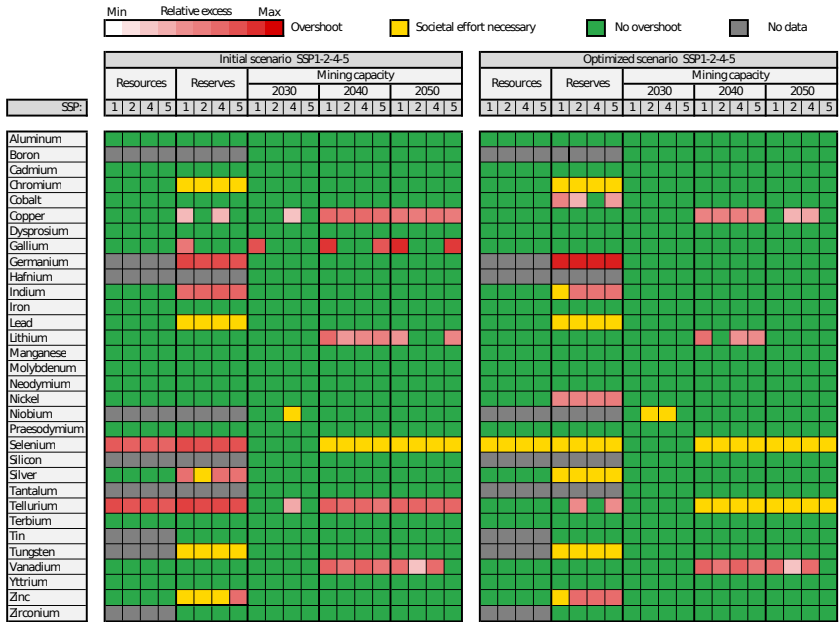


Fig. 6 Heatmap comparing metal availability risks under different supply constraints in the initial scenario (left) and the optimized scenario (right), with metal demand from the rest of the economy proportional to population growth. Each constraint is represented by four columns, corresponding to the availability assessment for SSP 1, 2, 4, and 5 under RCP 2.6. A gray square indicates missing data on resources or reserves for a given metal. A green square signifies that demand does not exceed the metal constraint. A yellow square denotes constraint overshoot, with more than 90% of the demand coming from the rest of the economy. A red square indicates that metal demand surpasses the constraints. The relative excess scale is calculated with a logarithmic normalization.

2 Supplementary Discussion

2.1 Regional specificity for power generation installation

Our model did not consider all regional specificity in technology deployment. Land availability was only considered for biomass, despite also being a significant constraint for CSP, solar photovoltaic (PV), and wind. Similarly, regional bio-resource availability was only factored in for hydroelectricity. By minimizing the difference between the initial and optimized scenarios for all five regions embedded in the IAM projections (ASIA, LAM, MAF, OECD, REF), we respected some regional specificity from IAMs, but with limited granularity. Moreover, we assumed a globalized metal supply, thereby excluding geopolitical supply risks [6].

To improve regional specificity, future research should apply the optimization model to national energy mixes. This would allow for the integration of region-specific constraints on technology deployment and metal supply to the model, supporting more targeted policymaking.

3 Supplementary Methods

3.1 Metal intensities of energy sources

For all the sub-technologies, we conducted a comprehensive literature review on metal intensities. The collected data are provided in Supplementary Data 1, sheet : MI. For each technology, the metal intensities (expressed in tons per GW) are summarized in a table. The “dataset” column indicates the number of literature sources compared. The “Min” and “Max” values represent the minimum and maximum metal intensities found, highlighting the variability within the literature for each data. Finally, the selected metal intensity value is given in the “MI” column, along with its corresponding reference and the hypothesis behind the choice.

To convey data reliability, the metal intensity values are color-coded:

- Blue: based on a single data point,
- Green: supported by multiple data points of the same order of magnitude,
- Yellow: variability in the dataset of approximately one order of magnitude (10^1),
- Red: substantial variability exceeding two orders of magnitude (10^2).

While comparing the data, Open LCA analysis, based on Ecoinvent metal intensities, are considered uncertain, since they have a broader framework than metal intensities from the literature.

3.1.1 Solar photovoltaic sub-technologies

Table 6 Metal intensities identical for all photovoltaic sub-technologies

Metal	Dataset	Min	Max	MI	Ref	Hypothesis
Aluminium	11	7 288	133 000	10 953	(Liang <i>et al.</i> , 2022)	Most data in the same order of magnitude. Most recent and globalized value is taken.
Chromium	1	-	-	3 761	(Watari <i>et al.</i> , 2019)	Only available data.
Molybdenum	1	-	-	200	(Watari <i>et al.</i> , 2019)	Only available data.
Iron	4	65 137	1 174 320	66 447	(European Commission, 2020)	Most values are around the same order of magnitude, except the highest one. The most recent and representative of the dataset is taken.
Tin	3	10	577	332	(Watari <i>et al.</i> , 2019)	Two values are in the same order of magnitude. The most representative of the dataset is taken.
Zinc	4	30	1 400	1 400	(Watari <i>et al.</i> , 2019)	Various studies with different order of magnitude in the dataset. The most recent data is chosen.

Table 7 Metal intensity of the c-Si solar technology

Metal	Dataset	Min	Max	MI	Ref	Hypothesis
Copper	12	250	4 640	3 765	(Watari <i>et al.</i> , 2019)	Variability in the dataset. The most recent and representative value of the dataset is taken.
Lead	7	6	550	39	(Watari <i>et al.</i> , 2019)	Variability in the dataset, but several studies in the same order of magnitude. The most recent and representative value of the dataset is taken.
Nickel	4	1	1 800	1.3	(Liang <i>et al.</i> , 2022)	Most values are close to 1, except the one from OpenLCA.
Silicium	5	3 948	6 630	4 000	(European Commission, 2020)	All values are around the same order of magnitude. The most recent one is taken.
Silver	24	10	84	30	(Watari <i>et al.</i> , 2019)	All values are around the same order of magnitude. The most representative of the dataset, and recent one is taken.

Table 8 Metal intensity of the thin-film CdTe solar technology

Metal	Dataset	Min	Max	MI	Ref	Hypothesis
Cadmium	3	50	138	87	(Watari <i>et al.</i> , 2019)	All data were in the same order of magnitude. The most representative of the dataset was taken.
Copper	2	3 091	5 181	3 091	(Watari <i>et al.</i> , 2019)	All data were in the same order of magnitude. The most recent one is chosen.
Lead	1	-	-	8	(Watari <i>et al.</i> , 2019)	Only available data.
Nickel	1	-	-	16	(Watari <i>et al.</i> , 2019)	Only available data.
Tellurium	7	15.2	156	92	(Watari <i>et al.</i> , 2019)	Most data were in the same order of magnitude. The most representative of the dataset is taken.

Table 9 Metal intensity of the CIGS solar technology

Metal	Dataset	Min	Max	MI	Ref	Hypothesis
Copper	6	20	4640	3 998	(Watari <i>et al.</i> , 2019)	The panel data from Watari is added to the specific one of CIGS.
Gallium	5	4	19	9	(Tokimatsu <i>et al.</i> , 2017)	All data around the same order of magnitude. The most representative one is taken.
Indium	5	13	31	28	(Tokimatsu <i>et al.</i> , 2017)	All data around the same order of magnitude. The most representative one is taken.
Selenium	5	35	161	61	(Watari <i>et al.</i> , 2019)	All data around the same order of magnitude. The most representative one is taken.

Table 10 Metal intensity of the a-SiGe solar technology

Metal	Dataset	Min	Max	MI	Ref	Hypothesis
Copper	1	-	-	4 710	(Månberger, 2018)	Only available data.
Germanium	1	-	-	48	(Månberger, 2018)	Only available data.
Indium	1	-	-	5	(Månberger, 2018)	Only available data.
Silicium	1	-	-	150	(European Commission, 2020)	Only available data.

3.1.2 Concentrated solar power sub-technologies

3.1.3 Wind sub-technologies

For all wind sub-technologies, the dataset from the European Commission (2020) is prioritized, as it is the most recent and technologically precise source available [7]. Numerous studies have collected metal intensities for wind technologies. The reported values are globally of the same order of magnitude, which supports their reliability.

For the GB-DFIG technology, the reported metal intensities are compared with values from other aggregated wind technologies and onshore wind, since DFIG systems are generally implemented in onshore settings.

Table 11 Metal intensity of Wind GB-DFIG

Metal	Dataset	Min	Max	MI	Ref
Aluminium	6	372	13 200	1 400	(European Commission, 2020)
Chromium	4	470	2 003	470	(European Commission, 2020)
Copper	8	1 400	9 621	1 400	(European Commission, 2020)
Dysprosium	2	0	2	2	(European Commission, 2020)
Iron	5	18 000	182 423	18 000	(European Commission, 2020)
Manganese	4	57	969	780	(European Commission, 2020)
Molybdenum	3	99	335	99	(European Commission, 2020)
Neodymium	1	–	–	12	(European Commission, 2020)
Nickel	4	377	1 490	430	(European Commission, 2020)
Praesodymium	2	0	3	0	(European Commission, 2020)
Terbium	2	0	2	0	(European Commission, 2020)
Zinc	4	330	5 500	5 500	(European Commission, 2020)

Similarly for the DD-EESG technology, the reported metal intensities are compared with values from other aggregated wind technologies and onshore wind, since EESG systems are generally implemented in onshore settings.

Table 12 Metal intensity of Wind DD-EESG technology

Metal	Dataset	Min	Max	MI	Ref
Aluminium	4	372	1 372	700	(European Commission, 2020)
Chromium	3	470	683	525	(European Commission, 2020)
Copper	5	2 000	5 000	5 000	(European Commission, 2020)
Dysprosium	3	0	6	6	(European Commission, 2020)
Iron	5	119 985	182 423	149 988	(European Commission, 2020)
Manganese	4	57	969	790	(European Commission, 2020)
Molybdenum	3	99	335	109	(European Commission, 2020)
Neodymium	1	-	-	28	(European Commission, 2020)
Nickel	3	340	403	340	(European Commission, 2020)
Praesodymium	2	3	9	9	(European Commission, 2020)
Terbium	2	1	1	1	(European Commission, 2020)
Zinc	3	5 450	5 500	5 500	(European Commission, 2020)

For the DD-PMSG technology, metal intensities are compared with other aggregated wind data, direct-drive and offshore wind, since DD-PMSG is generally offshore.

Table 13 Metal intensity of DD-PMSG

Metal	Dataset	Min	Max	MI	Ref
Aluminium	5	447	1 312	500	(European Commission, 2020)
Boron	3	6	7	6	(European Commission, 2020)
Chromium	3	372	525	525	(European Commission, 2020)
Copper	5	2 681	9 371	3 000	(European Commission, 2020)
Dysprosium	4	13	33	17	(European Commission, 2020)
Iron	5	106 765	443 772	126 933	(European Commission, 2020)
Manganese	4	57	1 028	790	(European Commission, 2020)
Molybdenum	3	99	335	109	(European Commission, 2020)
Neodymium	5	148	200	180	(European Commission, 2020)
Nickel	4	200	427	240	(European Commission, 2020)
Praesodymium	3	33	35	35	(European Commission, 2020)
Terbium	3	7	8.5	7	(European Commission, 2020)
Zinc	5	1 244	5 500	5 500	(European Commission, 2020)

To our knowledge, no other study than the one from the European Commission (2020) have specifically studied the hybrid GB-PMSG. Though, the REE metal intensities are found to be coherent since they are in between the ones from DD-PMSG and GB DFIG.

Table 14 Metal intensity of Wind GB-PMSG technology

Metal	Dataset	MI	Ref
Aluminium	1	1 600	(European Commission, 2020)
Boron	1	1	(European Commission, 2020)
Chromium	1	580	(European Commission, 2020)
Copper	1	950	(European Commission, 2020)
Dysprosium	1	6	(European Commission, 2020)
Iron	1	126 088	(European Commission, 2020)
Manganese	1	800	(European Commission, 2020)
Molybdenum	1	119	(European Commission, 2020)
Neodymium	1	51	(European Commission, 2020)
Nickel	1	440	(European Commission, 2020)
Praesodymium	1	4	(European Commission, 2020)
Terbium	1	1	(European Commission, 2020)
Zinc	1	5 500	(European Commission, 2020)

967 **3.1.4 Nuclear sub-technologies**

968 Fewer studies have explored the metal intensity required for nuclear sub-technologies,
969 leading to several data being chosen based on only one data point, and an important
970 variability between sources for some metal intensities. For specific sub-technologies, we
971 rely on the estimates from Dong et al. (2025), as this is the most recent and detailed
972 dataset available to our knowledge [8]. For pressurized water reactors, the reported
973 metal intensities are additionally compared with aggregated nuclear estimates, given
974 that PWRs remain the most widely deployed sub-technology today.
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Table 15 Metal intensities identical for all nuclear sub-technologies

Metal	Dataset	Min	Max	MI	Ref	Hypothesis
Aluminum	3	542	61 104	61 104	(Dones, R. <i>et al.</i> , 2007)	Important variability in the data. Open LCA data was excluded.
Chromium	4	427	5 444	2 190	(IEA, 2022)	Variability in data. Data from IEA was chosen, considered the most reliable source
Copper	10	25	2 500	1 473	(IEA, 2022)	Data from IEA was chosen, considered the most reliable source. All the data are around the same order of magnitude.
Iron	4	492	476 187	476 187	(Moss <i>et al.</i> , 2013)	Most recent data is chosen. Great variability in data.
Lead	3	4	20	4	(Moss <i>et al.</i> , 2013)	Use of Moss and al, coherent with Open LCA data.
Manganese	3	148	264	148	(IEA, 2022)	All the data are in the same order of magnitude. Data from IEA was chosen, considered the most reliable source.
Molybdenum	6	2.5	71	71	(IEA, 2022)	Data from IEA is considered the most reliable. Almost all data are in the same order of magnitude.
Nickel	8	256	3 355	1 297	(IEA, 2022)	Data from IEA was chosen, considered the most reliable source, and representative of the dataset.
Tin	2	0.3	5	5	(Moss <i>et al.</i> , 2013)	Use of Moss and al, coherent with Open LCA data.
Tungsten	1	-	-	5	(Moss <i>et al.</i> , 2013)	Only available data.
Yttrium	1	-	-	1	(Moss <i>et al.</i> , 2013)	Only available data.
Zinc	1	-	-	48	(Open LCA Analysis)	Only available data.
Zirconium	2	14	31	31	(Moss <i>et al.</i> , 2013)	Use of Moss and al, coherent with Open LCA data.

Table 16 Metal intensity of the PWR technology

Metal	Dataset	Min	Max	MI	Ref	Hypothesis
Cadmium	1	-	-	0.4	(Dong <i>et al.</i> , 2025)	Only available data
Hafnium	3	0.5	1	0.5	(Dong <i>et al.</i> , 2025)	Most recent data is chosen.
Indium	3	1.4	2	1.4	(Dong <i>et al.</i> , 2025)	Most recent and precise data was chosen. Same order of magnitude.
Niobium	2	0.3	2	0.3	(Dong <i>et al.</i> , 2025)	The most recent data is chosen.
Silver	3	7.5	8.3	7.3	(Dong <i>et al.</i> , 2025)	All data are in the same order of magnitude, the most recent data is chosen.
Vanadium	2	1	3	3	(Dong <i>et al.</i> , 2025)	Most recent and precise data was chosen. Same order of magnitude.

Table 17 Metal intensity of the FBR technology

Metal	Dataset	Min	Max	MI	Ref	Hypothesis
Niobium	1	-	-	0.01	(Dong <i>et al.</i> , 2025)	Only available data.
Silver	1	-	-	0	(Dong <i>et al.</i> , 2025)	Only available data.
Vanadium	1	-	-	1.7	(Dong <i>et al.</i> , 2025)	Only available data.

Table 18 Metal intensity of the HTGR technology

Metal	Dataset	Min	Max	MI	Ref	Hypothesis
Niobium	1	-	-	0	(Dong <i>et al.</i> , 2025)	Only available data.
Silver	1	-	-	0	(Dong <i>et al.</i> , 2025)	Only available data.
Vanadium	1	-	-	1.2	(Dong <i>et al.</i> , 2025)	Only available data.

3.1.5 Hydroelectricity

In general, few studies report metal intensities for hydroelectricity. Most values for this technology are based on the dataset from Ashby (2013), which provides the most comprehensive coverage [9]. Data from Open LCA and Ecoinvent are used for comparison, to assess the reliability of the reported intensities [10].

Table 19 Metal intensity of hydroelectricity

Metal	Dataset	Min	Max	MI	Ref	Hypothesis
Aluminum	2	1 635	3 400	3 400	(Ashby, 2013)	Use of Ashby, coherent with Open LCA data.
Chromium	2	1 500	12 393	1 500	(Ashby, 2013)	Use of Ashby, coherent with Open LCA data.
Copper	4	1 050	6 735	3 160	(Lea A. Copper Association, 2016)	The copper association data is considered the most reliable, all data are in the same order of magnitude.
Iron	2	175 000	338 665	222 090	(Ashby, 2013)	Use of Ashby, coherent with Open LCA data.
Lead	2	59	300	300	(Ashby, 2013)	Use of Ashby, coherent with Open LCA data.
Magnesium	2	37	100	100	(Ashby, 2013)	Use of Ashby, coherent with Open LCA data.
Manganese	2	200	3 393	200	(Ashby, 2013)	Use of Ashby, coherent with Open LCA data.
Molybdenum	2	250	330	330	(Dong <i>et al.</i> , 2025)	Most recent data is chosen.
Zinc	2	253	400	400	(Ashby, 2013)	Use of Ashby, coherent with Open LCA data.

3.1.6 Biomass

In general, few metal intensities were found for biomass energy. Similarly to hydroelectricity, we mostly used the dataset from Ashby (2013) [9]. An Open LCA analysis is used as a comparison, but since it only studies wood energy, it is considered uncertain [10].

Table 20 Metal intensity of biomass

Metal	Dataset	Min	Max	MI	Ref	Hypothesis
Aluminum	4	184	3 900	3 900	(Ashby, 2013)	Since two studies had the same order of magnitude, they were prioritized. The most recent value was taken. Variability in the dataset.
Chromium	2	2	269	2	(Ashby, 2013)	The aggregated data on all biomass methods is chosen.
Cobalt	2	0.07	2	2	(Ashby, 2013)	The aggregated data on all biomass methods is chosen. Coherent with Open LCA data.
Copper	4	34	2 270	2 175	(Lea A., Copper Association, 2016)	Two data are similar, and the copper association data is considered the most reliable. Variability in the dataset.
Iron	4	33 595	135 047	52 000	(Sullivan <i>et al.</i> , 2010)	Use of the data which represents most the dataset, variability in the dataset.
Lead	2	4	104	104	(Ashby, 2013)	The aggregated data on all biomass methods is chosen, variability in the dataset.
Nickel	2	20	1 597	20	(Ashby, 2013)	The aggregated data on all biomass methods is chosen, variability in the dataset.
Zinc	2	18	160	160	(Ashby, 2013)	The aggregated data on all biomass methods is chosen, variability in the dataset.

3.1.7 Geothermal energy

In general, few metal intensities were found for geothermal energy. We mostly used Ashby (2013) and Moss *et al.* (2011) dataset which was the most complete ones. We compared it with the Open LCA data for coherence.

Table 21 Metal intensity of geothermal energy

Metal	Dataset	Min	Max	MI	Ref	Hypothesis
Aluminum	3	2 386	46 100	2 386	(Karlsdóttir <i>et al.</i> , 2015)	Use of the most recent data, which is around the same order of magnitude than Open LCA.
Chromium	2	13 663	64 405	64 405	(Ashby, 2013)	Use of Ashby, coherent with Open LCA data.
Copper	5	597	3 605	2 175	(Lea A., Copper Association, 2016)	The copper association data is considered the most reliable. It has a value similar to all the data found, except the minimum one.
Iron	4	133 513	800 495	579 527	(Ashby, 2013)	All data are in the same order of magnitude. The most representative of the dataset was chosen. The Open LCA data was excluded since its framework is wider than ours.
Manganese	2	4 325	9 455	4 325	(Ashby, 2013)	Use of Ashby, coherent with Open LCA data.
Molybdenum	1	-	-	7 209	(Moss <i>et al.</i> , 2011)	Only available data
Nickel	2	12 023	120 155	120 155	(Ashby, 2013)	Use of Ashby, coherent with Open LCA data, but with some variability in the dataset.
Niobium	1	-	-	128	(Moss <i>et al.</i> , 2011)	Only available data
Tantalum	2	0.10	64	64	(Moss <i>et al.</i> , 2011)	Use of Moss and al, coherent with Open LCA data, but with some variability in the dataset.

3.1.8 Fossil sub-technologies

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1247 **Table 22** Metal intensity of coal

1248	Metal	Dataset	Min	Max	MI	Ref	Hypothesis
1249	Aluminium	2	500	1 033	1 033	(Van Oorschot <i>et al.</i> , 2022)	Variability in data. Use the only specific one for this sub-technology.
1250	Chromium	1	-	-	307	(IEA, 2022)	Only available data
1251	Cobalt	1	-	-	201	(IEA, 2022)	Only available data
1252	Copper	4	90	1 510	1 150	(IEA, 2022)	3 data are around the same order of magnitude. The IEA data is considered the most reliable.
1253	Manganese	1	-	-	5	(IEA, 2022)	Only available data
1254	Nickel	1	-	-	721	(IEA, 2022)	Only available data
1255	Iron	2	52 500	77 035	75 387	(Van Oorschot <i>et al.</i> , 2022)	Use the only data specific for this sub-technology.

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1261 **Table 23** Metal intensity of hydrogen

1270	Metal	Dataset	Min	Max	MI	Ref	Hypothesis
1271	Aluminium	1	-	-	922	(Van Oorschot <i>et al.</i> , 2022)	Only available data
1272	Copper	2	922	1 110	1 110	(Lea A., Copper Association, 2016)	The copper association data is considered the most reliable. It has a value similar to all the data found, except the minimum one.
1273	Iron	1	-	-	15 206	(Van Oorschot <i>et al.</i> , 2022)	Only available data.

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Table 24 Metal intensity of natural gas

Metal	Dataset	Min	Max	MI	Ref	Hypothesis
Aluminium	3	500	922	922	(Van Oorschot <i>et al.</i> , 2022)	Use the only data specific for this sub-technology.
Chromium	2	48	57	48	(IEA, 2022)	Same order of magnitude. The IEA data is considered the most reliable.
Copper	4	90	1 110	1 100	(IEA, 2022)	Most data are in the same order of magnitude. The IEA data is considered the most reliable.
Iron	1	-	-	6 341	Open LCA Analysis	Only available data
Lead	1	-	-	5	Open LCA Analysis	Only available data
Magnesium	1	-	-	2	Open LCA Analysis	Only available data
Manganese	1	-	-	27	Open LCA Analysis	Only available data
Nickel	2	16	49	16	(IEA, 2022)	Same order of magnitude. The IEA data is considered the most reliable.
Zinc	1	-	-	20	Open LCA Analysis	Only available data

Table 25 Metal intensity of oil

Metal	Dataset	Min	Max	MI	Ref	Hypothesis
Aluminium	1	-	-	518	Open LCA Analysis	Only available data
Chromium	1	-	-	904	Open LCA Analysis	Only available data
Copper	2	1 100	1 513	1 100	(Lea A., Copper Association, 2016)	The copper association data is considered the most reliable.
Iron	1	-	-	78 791	Open LCA Analysis	Only available data
Lead	1	-	-	9	Open LCA Analysis	Only available data
Manganese	1	-	-	359	Open LCA Analysis	Only available data
Nickel	1	-	-	676	Open LCA Analysis	Only available data
Zinc	1	-	-	40	Open LCA Analysis	Only available data

3.2 Metal intensities of electric vehicles

The metal intensity tables for vehicles are similar than the ones for energy sources. Though, since the metal intensities are given per kWh or vehicle, the color code on variability is adapted:

- Blue: based on a single data point,
- Green: supported by multiple data points of the same order of magnitude,
- Yellow: variability in the dataset above 200,
- Red: variability in the dataset above 500.

3.2.1 Vehicle body

Metal intensities of the vehicle body are reported in g/vehicle. Since the metals used in the vehicle body are not considered potentially critical for supply, we did not perform an extensive literature comparison. We selected the metal intensities from Watari *et al.*, as these were, to our knowledge, the only available data providing detailed information on specific EV sub-technologies [11].

Table 26 Metal intensity of a BEV vehicle body

Metal	MI	Ref	Hypothesis
Aluminium	127 302	(Watari <i>et al.</i> , 2019; Fishman <i>et al.</i> , 2018)	Only available data
Chromium	11 850	(Watari <i>et al.</i> , 2019)	Only available data
Iron	902 171	(Watari <i>et al.</i> , 2019; Fishman <i>et al.</i> , 2018)	Only available data
Lead	310	(Watari <i>et al.</i> , 2019)	Only available data
Zinc	100	(Watari <i>et al.</i> , 2019)	Only available data

Table 27 Metal intensity of an ICEV vehicle body

Metal	MI	Ref	Hypothesis
Aluminium	71 300	(Watari <i>et al.</i> , 2019; Fishman <i>et al.</i> , 2018)	Only available data
Chromium	0	(Watari <i>et al.</i> , 2019)	Only available data
Iron	902 171	(Watari <i>et al.</i> , 2019; Fishman <i>et al.</i> , 2018)	Only available data
Lead	300	(Watari <i>et al.</i> , 2019)	Only available data
Zinc	100	(Watari <i>et al.</i> , 2019)	Only available data

Table 28 Metal intensity of vehicle a PHEV body

Metal	MI	Ref	Hypothesis
Aluminium	110 658	(Watari <i>et al.</i> , 2019; Fishman <i>et al.</i> , 2018)	Only available data
Chromium	12 789	(Watari <i>et al.</i> , 2019)	Only available data
Iron	902 171	(Watari <i>et al.</i> , 2019; Fishman <i>et al.</i> , 2018)	Only available data
Lead	320	(Watari <i>et al.</i> , 2019)	Only available data
Zinc	100	(Watari <i>et al.</i> , 2019)	Only available data

3.2.2 Vehicle motor

Metal intensities of the vehicle motor are given in g/vehicle. Some studies gave metal intensities in g/kW. In order to compare the different metal intensities of the motors, we made the hypothesis of a mean engine power at 70 kW [12].

The metal intensities are given for a usual motor, using copper wires. In the study, we modeled alternative permanent magnet and induction motors using aluminum wires. We supposed a substitution of 3.4 tons of copper by 1 ton of aluminum, following the substitution potential of Ayat *et al.* study, for the wire II [13].

Table 29 Metal intensity of a permanent magnet motor

Metal	Dataset	Min	Max	MI	Ref	Hypothesis
Aluminium	1	-	-	0	(Ayat <i>et al.</i> , 2017)	Only available data
Copper	1	-	-	13 885	(Månberger & Stenqvist, 2018)	Only available data, supposed a motor of 70 kW
Dysprosium	3	3.6	140	83	(Watari <i>et al.</i> , 2018)	Variability in dataset, used the most representative data
Neodymium	3	266	700	695	(Watari <i>et al.</i> , 2018)	Two data were similar. Used the most representative one
Praesodymium	1	-	-	1	(Watari <i>et al.</i> , 2018)	Only available data, supposed a motor of 70 kW

Table 30 Metal intensity of an induction motor

Metal	Dataset	Min	Max	MI	Ref	Hypothesis
Aluminium	1	-	-	0	(Ayat <i>et al.</i> , 2017)	Only available data
Copper	1	-	-	33 674	(Månberger & Stenqvist, 2018)	Only available data, supposed a motor of 70 kW
Dysprosium	2	0	0	0	(Watari <i>et al.</i> , 2018)	All data were identical
Neodymium	2	0	0	0	(Watari <i>et al.</i> , 2018)	All data were identical
Praesodymium	1	-	-	0	(Watari <i>et al.</i> , 2018)	Only available data, supposed a motor of 70 kW

3.2.3 Vehicle battery

The metal intensities of batteries are given in g/kWh, in order to account for the variation in battery capacities between vehicle types (EV, PHEV and HyEV).

In order to maintain a coherence between the battery capacity metal intensities, we used the data from IRENA for all batteries [14]. These data were the most recent and precise data in terms of sub-technologies available to our knowledge. In order to test the variability in dataset, we compared it to various studies. We noted a certain variability in the data. Notably, Ecoinvent data were usually quite different from other dataset [10], and IRENA estimation for lithium was usually superior than for other studies.

Table 31 Metal intensity of an LFP battery

Metal	Dataset	Min	Max	MI	Ref	Hypothesis
Cobalt	3	0	1 536	0	(IRENA, 2024)	Two datasets were at 0.
Copper	2	0	493	493	(IRENA, 2024)	IRENA's data is preferred to the one from ecoinvent.
Lithium	4	104	511	511	(IRENA, 2024)	The most recent data is chosen.
Manganese	2	0	1 492	0	(IRENA, 2024)	IRENA's data is preferred to the one from ecoinvent.
Nickel	2	0	140	0	(IRENA, 2024)	IRENA's data is preferred to the one from ecoinvent.

Table 32 Metal intensity of an LMFP battery

Metal	Dataset	Min	Max	MI	Ref	Hypothesis
Cobalt	1	-	-	0	(IRENA, 2024)	Only available data
Copper	1	-	-	484	(IRENA, 2024)	Only available data
Lithium	1	-	-	520	(IRENA, 2024)	Only available data
Manganese	1	-	-	317	(IRENA, 2024)	Only available data
Nickel	1	-	-	0	(IRENA, 2024)	Only available data

Table 33 Metal intensity of an LMO battery

Metal	Dataset	Min	Max	MI	Ref	Hypothesis
Cobalt	2	0	500	0	(IRENA, 2024)	Most recent data is chosen.
Copper	1	-	-	440	(IRENA, 2024)	Only available data
Lithium	2	113	599	599	(IRENA, 2024)	Most recent data is chosen.
Manganese	1	-	-	1 277	(IRENA, 2024)	Only available data
Nickel	1	-	-	0	(IRENA, 2024)	Only available data

Table 34 Metal intensity of a Na-ion battery

Metal	Dataset	Min	Max	MI	Ref	Hypothesis
Cobalt	2	0	160	0	(IRENA, 2024)	IRENA's data is preferred to the one from ecoinvent.
Copper	2	0	200	0	(IRENA, 2024)	IRENA's data is preferred to the one from ecoinvent.
Lithium	2	0	0	0	(IRENA, 2024)	Both data are identical.
Manganese	2	113	555	555	(IRENA, 2024)	IRENA's data is preferred to the one from ecoinvent.
Nickel	2	335	500	335	(IRENA, 2024)	IRENA's data is preferred to the one from ecoinvent.

Table 35 Metal intensity of an aggregated Gr-NMC battery

Metal	Dataset	Min	Max	MI	Ref	Hypothesis
Cobalt	2	132	300	132	(IRENA, 2024)	Most recent data is chosen.
Copper	1	-	-	317	(IRENA, 2024)	Only available data
Lithium	2	140	608	608	(IRENA, 2024)	Most recent data is chosen.
Manganese	1	-	-	159	(IRENA, 2024)	Only available data
Nickel	1	-	-	564	(IRENA, 2024)	Only available data

Table 36 Metal intensity of a Gr-NMC 433 battery

Metal	Dataset	Min	Max	MI	Ref	Hypothesis
Cobalt	2	138	305	138	(IRENA, 2024)	Most recent data is chosen.
Copper	1	-	-	310	(IRENA, 2024)	Only available data
Lithium	2	145	610	610	(IRENA, 2024)	Most recent data is chosen.
Manganese	1	-	-	160	(IRENA, 2024)	Only available data
Nickel	1	-	-	530	(IRENA, 2024)	Only available data

Table 37 Metal intensity of a Gr-NMC 532 battery

Metal	Dataset	Min	Max	MI	Ref	Hypothesis
Cobalt	3	212	2126	212	(IRENA, 2024)	Two data are similar. Most recent data is chosen.
Copper	2	317	628	317	(IRENA, 2024)	IRENA's data is preferred to the one from ecoinvent.
Lithium	3	150	655	655	(IRENA, 2024)	Data quite different. The most recent one is chosen.
Manganese	3	193	786	193	(IRENA, 2024)	Data quite different. The most recent one is chosen.
Nickel	3	494	1821	514	(IRENA, 2024)	Two data are similar. Most recent data is chosen.

Table 38 Metal intensity of a Gr-NMC 622 battery

Metal	Dataset	Min	Max	MI	Ref	Hypothesis
Cobalt	2	145	320	145	(IRENA, 2024)	Most recent data is chosen.
Copper	1	-	-	320	(IRENA, 2024)	Only available data
Lithium	2	150	615	615	(IRENA, 2024)	Most recent data is chosen.
Manganese	1	-	-	165	(IRENA, 2024)	Only available data
Nickel	1	-	-	550	(IRENA, 2024)	Only available data

Table 39 Metal intensity of a Gr-NMC 811 battery

Metal	Dataset	Min	Max	MI	Ref	Hypothesis
Cobalt	2	160	340	160	(IRENA, 2024)	Most recent data is chosen.
Copper	1	-	-	330	(IRENA, 2024)	Only available data
Lithium	2	155	620	620	(IRENA, 2024)	Most recent data is chosen.
Manganese	1	-	-	170	(IRENA, 2024)	Only available data
Nickel	1	-	-	580	(IRENA, 2024)	Only available data

Table 40 Metal intensity of Gr-NCA batteries (aggregated and sub-technologies)

Metal	Sub-technology	Dataset	Min	Max	MI	Ref	Hypothesis
Cobalt	Gr-NCA	4	106	920	105.7	(IRENA, 2024)	Most data are in the same order of magnitude, except the maximum one. The most recent one is chosen.
Cobalt	Gr-NCA85	1	-	-	105.8	(IRENA, 2024)	Only available data
Cobalt	Gr-NCA95	1	-	-	105.7	(IRENA, 2024)	Only available data
Copper	All	2	264	706	264	(IRENA, 2024)	IRENA's data is preferred to the one from ecoinvent.
Lithium	All	4	22	572	572	(IRENA, 2024)	Most recent data is chosen.
Manganese	All	3	0	59	0	(IRENA, 2024)	All data are in the same order of magnitude. Most recent data is chosen.
Nickel	Gr-NCA	3	705	950	705	(IRENA, 2024)	All data are in the same order of magnitude. Most recent data is chosen.
Nickel	Gr-NCA85	1	-	-	674	(IRENA, 2024)	Only available data
Nickel	Gr-NCA95	1	-	-	735	(IRENA, 2024)	Only available data

1703 **3.2.4 Vehicle battery capacity**

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1706 **Table 41** Battery capacities by sub-technology

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Vehicle type	Dataset	Min	Max	Battery capacity	Ref	Hypothesis
BEV	5	28	79.23	62.5	(Adamas Intelligence, 2023)	Chose the most recent and precise data in terms of electric vehicle.
HyEV	2	1.3	1.4	1.3	(Adamas Intelligence, 2023)	Chose the most recent and precise data in terms of electric vehicle.
PHEV	3	7	70.5	21.8	(Adamas Intelligence, 2023)	Chose the most recent and precise data in terms of electric vehicle.

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1723 **3.3 Scenario comparison of vehicle stock projections**

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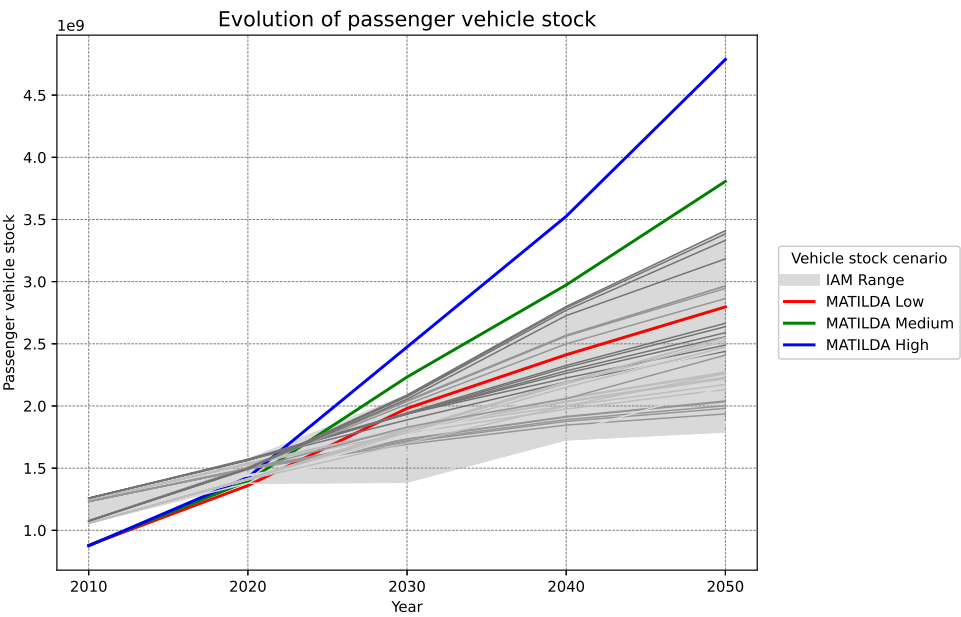
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1748



1746 **Fig. 7** Comparison of the passenger vehicle stock evolution of different IAM scenarios and Matilda scenarios.

3.4 Scenario mapping

The qualitative descriptions of SSP-RCP constraints were used to associate them with IEA scenarios for future EV projections, as well as storage and grid metal requirements, and recycling rate evolution. This mapping is summarized in the supplementary Tab.42. Although GDP growth assumptions are intrinsic to IAM generated SSP narratives [15], they were explicitly specified in the table because of their influence on metal demand growth from the rest of the economy.

IEA metal demand projections and EV requirements are based on three policy scenarios: the Net Zero Emissions by 2050 (NZE), the Announced Pledges Scenario (APS), and the Stated Policies Scenario (SPS) [16]. SSP1 and SSP4, characterized by a strong uptake of low-carbon energy sources and an increasing share of EVs, were linked to the NZE and APS scenarios. The NZE scenario outlines a pathway for the global energy sector to achieve net zero CO₂ emissions by 2050, aiming to limit global warming to 1.5°C, and was therefore mapped to RCP1.9. APS assumes that all announced long-term emissions and energy access targets will be achieved, aiming for a temperature rise of 1.7°C in 2100, and was linked to RCP2.6. STEPS, which reflects current policy settings, was linked to all other IAM SSP-RCP scenarios.

Regarding recycling rate evolution, we quantitatively translated the mapping proposed by Wang et al. between material-related strategies and shared socioeconomic pathways [17]. For SSP1, we globally extrapolated optimistic future recycling rates based on EU targets [18]. For SSP2, we assumed a neutral scenario with recycling rates 20% lower than those of SSP1. Finally, for SSP4 and SSP5, which represent more conservative pathways, we assumed that recycling rates would remain constant over time. Specific recycling rates are presented in Supplementary Data 2 (sheet: Recycling_Rate).

Table 42 Scenario mapping of SSP-RCP2.6 IAM scenarios

SSP-RCP scenario	SSP1-26	SSP2-26	SSP4-26	SSP5-26
GDP growth	2020-2040: 3% 2040-2050: 1.8%	2020-2040: 2.7% 2040-2050: 1.7%	2020-2040: 2.7% 2040-2050: 1.1%	2020-2040: 3.1% 2040-2050: 2.5%
IEA scenario	APS	STEPS	APS	STEPS
Recycling rate	Optimistic	Neutral	Constant	Constant

3.5 Interchangeability between sub-technologies

Changing one technology by another is penalized in the optimization code, since the objective function minimizes the differences between the initial and the optimized scenario. Though, when sub-technologies had similar use and were supposed easily interchangeable, we did not penalize the replacement of one sub-technology by the other. These assumptions are summarized in Tab.43.

In the next paragraphs, we present the technological choices made in terms of metal substitution and sub-technological interchangeability. All the abbreviations used for the technologies are described in the subsection 3.6.15, at the end of this document.

Table 43 Aggregated technologies and interchangeable sub-groups

Technological mix	Aggregated technologies	Interchangeable sub-technologies
Energy sources	Solar crystalline silicon	Sol C-Si – silver conductor Sol C-Si – copper conductor
	Solar thin-film	Solar CdTe Solar CIGS Solar a-SiGe
	CSP	CSP Ag-GaAs CSP Al-MgF2
	Wind	GB-DFIG DD-PMSG DD-EESG GB-PMSG
	Nuclear	PWR FBR HTGR
	Gr-NMC	Gr-NMC(111) Gr-NMC(433) Gr-NMC(532) Gr-NMC(622) Gr-NMC(811)
Electric vehicle batteries	Gr-NCA	Gr-NCA95 Gr-NCA85
Electric vehicle engines	PM	PM - copper wires PM - aluminum wires
	Ind	Ind - copper wires Ind - aluminum wires

3.5.1 Solar photovoltaic sub-technologies

The energy transition could rely on a massive expansion of solar power in the coming years [16]. The different solar photovoltaic (PV) sub-technologies require a wide variety

of metals, leading to significant fluctuations in demand depending on the market shares considered [12]. Exploring the substitution potential of these technologies could provide flexibility in the choice of sub-technology.

Silver substitution in crystalline silicon solar

Currently, the dominant solar PV technology is the crystalline silicon (c-Si) with silver conductors. Though, silver resources and mining capacity may not be sufficient to meet the growing demand in the solar sector [19], highlighting the need to find substitution solutions for this metal [20]. The recent metallization of silicon cell contacts with copper makes it possible to replace the use of silver in c-Si [19]. Since copper and silver have similar conductivity properties, this replacement would only marginally affect the efficiency of c-Si while maintaining the same uses. In this study, interchangeability between the use of silver and copper conductors was considered, with an equivalent mass substitution.

Interchangeability between solar photovoltaic sub-technologies

Contrary to c-Si, solar PV thin-film solar cells, which are lighter and more flexible, can be integrated directly into buildings or windows. However, they also require metals whose supply may be limited [21]. Since thin-film solar allows for more diverse PV applications and has a slightly lower efficiency than c-Si solar, complete interchangeability between these technologies cannot be assumed. However, by using different metals, thin-film technologies are declined in various variants: Copper Indium Gallium Selenide (CIGS), Cadmium Telluride (CdTe) and Amorphous Silicon Germanium (a-SiGe). Since thin-film variants have similar use and efficiencies, they were considered interchangeable in this study.

3.5.2 Concentrated solar power sub-technologies

Substitution in concentrated solar technologies

CSP technologies use silver and gallium as solar reflectors. Since these metals could be limited by resources and mining capacity, substitution strategies are studied. Theoretically, it has been shown that the silver reflector could be replaced by aluminum, by combining it to a MgF₂ (Magnesium fluoride) layer rather than the generally used GaAs (Gallium Arsenide) one [22]. Since this substitution would lead to an even greater solar reflectance, and therefore no efficiency losses, the substitution potential was included in the optimization mix. A mass-equivalent ratio was applied. As aluminum and magnesium were not constrained by supply in the model, the uncertainty related to the mass ratio did not affect the results.

Interchangeability between CSP sub-technologies

CSP tower and parabolic require relatively similar types of metals, although the quantity varies depending on the technology variant [11, 12, 23]. Additionally, CSP sub-technologies have different efficiencies and require different installation spaces: tower CSP is deployed at height, while parabolic CSP requires a large floor space. Therefore, we did not model an interchangeability between CSP sub-technologies.

1887 **3.5.3 Wind sub-technologies**

1888

1889

1890 Projections of the future energy mix point to an accelerated expansion of wind power,
1891 with a particular growth in wind turbines [16]. Wind turbines with permanent magnets
1892 are among the growing technologies, due to their advantages for use [7, 16]. However,
1893 these generators rely on rare earth elements (REEs), likely to be constrained in the
1894 coming years [3]. The substitution potential of this technology is therefore analyzed.

1895

1896 *Substitution of rare earth elements in permanent magnets for wind* 1897 *turbines*

1898 Substituting the metals used in permanent magnets with more abundant metals has
1899 been explored in the laboratory. Although Pathak et al. studied the possibility of
1900 replacing NdFeB magnets with cobalt and cerium, this solution was deemed unlikely
1901 [24]. Terbium could potentially replace dysprosium, but this metal also poses supply
1902 risks. Ultimately, based on current knowledge, experts agree that permanent magnet
1903 metals are unlikely to be substituted in the short term. [25]

1904

1905 *Interchangeability between wind turbine sub-technologies*

1906 Several wind sub-technologies are already used. Direct-Drive Permanent Magnet Syn-
1907 chronous Generator (DD-PMSG) technology offers a real advantage for applications,
1908 both in terms of efficiency and cost, particularly thanks to a significant reduction in
1909 maintenance requirements. Therefore, it would not be justified to consider interchange-
1910 ability between wind technologies in our study. However, DD-PMSG wind turbines are
1911 also expected to be deployed on land in the future [26]. Alternatives such as GB-SCIG
1912 & DFIG, DD-EESG, and GM-PMSG technologies, already in use on land offer viable
1913 solutions with lower rare earth requirements [25]. Therefore, this study considers full
1914 flexibility between wind technology variants.

1915

1916 **3.5.4 Nuclear sub-technologies**

1917

1918 Current nuclear is mainly composed of Pressurized Water Reactors (PWR), which
1919 among others rely on restricted silver metal. Other reactors, such as the High-
1920 temperature Gas-Cooled Reactor (HTGR) and the Fast Breeder Reactor (FBR) do
1921 not rely on silver component. The HTGR and FBR are expected to be ready for mas-
1922 sive deployment after 2030 [8, 27]. Since they offer many advantages in comparison to
1923 the PWR reactor (reduced uranium requirements, more economical, enhanced safety,
1924 minimal waste) the nuclear sub-technologies were considered interchangeable.

1925

1926 **3.5.5 Other energy sources**

1927

1928 For hydroelectricity, even though different sub-technologies exist, they require different
1929 regional bio-resource availability and have only small variations in metal intensities
1930 [8]. Similarly, different biomass energy sources require very similar metal intensities
1931 [8]. Thus, they were excluded from the scope of this study.

1932

3.5.6 Electric vehicle battery chemistries	1933
Several battery chemistries exist for EVs, using different metals for the anode and cathode. However, within a specific chemistry, some sub-technologies differ only in the ratio of each metal. Since these technologies have very similar construction processes and use, they were considered interchangeable [14].	1934 1935 1936 1937 1938
3.5.7 Electric vehicle engines	1939 1940
EV conductors are composed of copper, which could be constrained by mining capacity in the future. Though, studies have shown that 60% of current copper use could be replaced [28]. Specifically, the substitution of copper by aluminum as a conductor in EV motors has been proven achievable, with marginal efficiency loss, reduced cost and weight [13]. Thus, the motors using copper or aluminum as wires were supposed interchangeable, with a mass ratio derived from the conductor II of the 2017 Ayat and al. study.	1941 1942 1943 1944 1945 1946 1947 1948 1949 1950 1951 1952 1953 1954 1955 1956 1957 1958 1959 1960 1961 1962 1963 1964 1965 1966 1967 1968 1969 1970 1971 1972 1973 1974 1975 1976 1977 1978

1979 3.6 Optimization code

1980 The optimization code was carried out using the Pyomo environment, with a CPLEX
1981 solver. It is open source and can be found in a Zenodo repository [29].
1982

1983 3.6.1 Sets 1984

1985 The variables and parameters used in this code depend on specific sets over which the
1986 indices vary. These sets are presented in this section. All the abbreviations used for
1987 the technologies are described in the subsection 3.6.15 at the end of this document.
1988

1989 *Sets defined by the scope of the study*

1990 Consider the set of metals M , defined as:

1991
1992 $M = \{\text{Aluminum, Boron, Cadmium, Chromium, Cobalt, Copper,}$
1993 $\text{Dysprosium, Gallium, Germanium, Hafnium, Indium, Iron, Lead,}$
1994 $\text{Molybdenum, Neodymium, Nickel, Niobium, Praesodymium, Selenium,}$
1995 $\text{Silicon, Silver, Tantalum, Tellurium, Terbium, Tin, Vanadium,}$
1996 $\text{Yttrium, Zinc, Zirconium}\}, \quad m \in M.$
1997
1998

1999 Consider the set of decades D , defined as:

2000
2001 $D = \{2020, 2030, 2040, 2050\}, \quad d \in D.$
2002

2003 Consider the set of years Y , defined as:

2004
2005 $Y = \{2018, 2019, \dots, 2050\}, \quad y \in Y.$
2006

2007 Consider the set of regions R , defined as:

2008
2009 $R = \{\text{ASIA, LAM, MAF, OECD, REF}\}, \quad r \in R.$
2010

2011 **ASIA**: Most Asian countries except Middle East, Japan, and former Soviet Union.

2012 **LAM**: Latin America and Caribbean countries.

2013 **MAF**: Middle East and African countries.

2014 **OECD**: OECD member and candidate countries of the 90s and EU.

2015 **REF**: Countries of Eastern Europe and former Soviet Union undergoing economic
2016 reform.
2017

2018

2019

2020

2021

2022

2023

2024

Sets and subsets of energy generation technologies	2025
Consider the set of energy generation technologies T , defined as:	2026
$T = \{\text{Sol_C-si_Silver}, \text{Sol_C-si_Copper}, \text{Sol_CdTe}, \text{Sol_CIGS}, \text{Sol_a-SiGe},$	2027
$\text{Sol_CSP_parabol_Al_MgF2}, \text{Sol_CSP_tower_Al_MgF2}, \text{Sol_CSP_parabol_Ag_GaAs}, \text{Sol_CSP_tower_Ag_GaAs},$	2028
$\text{Wind_DD-EESG_}, \text{Wind_GB-DFIG_SCIG_}, \text{Wind_DD-PMSG_}, \text{Wind_GB-PMSG_},$	2029
$\text{Wind_DD-EESG}, \text{Wind_GB-DFIG_SCIG}, \text{Wind_DD-PMSG}, \text{Wind_GB-PMSG},$	2030
$\text{Nuclear_PWR}, \text{Nuclear_HTGR}, \text{Nuclear_FBR}, \text{Hydro}, \text{Biomass}, \text{Geothermal},$	2031
$\text{Foss_Coal}, \text{Foss_NaturalGas}, \text{Foss_Oil}, \text{Foss_Hydrogen}\}, \quad t \in T.$	2032
	2033
In order to add flexibility between some sub-technologies considered inter-	2034
changeable, an other set of energy generation technologies is defined, with the	2035
sub-technologies aggregated in technological groups.	2036
	2037
Consider the set of aggregated energy generation technologies T_{agg} , defined as:	2038
	2039
$T_{\text{agg}} = \{\text{Sol_C-si}, \text{Sol_Thin_Film}, \text{Sol_CSP_parabol}, \text{Sol_CSP_tower},$	2040
$\text{Wind_}, \text{Wind_DD-EESG}, \text{Wind_GB-DFIG_SCIG},$	2041
$\text{Wind_DD-PMSG}, \text{Wind_GB-PMSG}, \text{Nuclear}, \text{Hydro},$	2042
$\text{Biomass}, \text{Geothermal}, \text{Foss_Coal}, \text{Foss_NaturalGas},$	2043
$\text{Foss_Oil}, \text{Foss_Hydrogen}\}, \quad t_{\text{agg}} \in T_{\text{agg}}.$	2044
	2045
	2046
	2047
Consider the subset of fossil technologies T_{foss} , defined as:	2048
	2049
$T_{\text{foss}} = \{\text{Foss_Coal}, \text{Foss_NaturalGas}, \text{Foss_Oil}, \text{Foss_Hydrogen}\}, \quad t_{\text{foss}} \in T_{\text{foss}}.$	2050
	2051
	2052
Sets and subsets of vehicles	2053
The set of vehicles V , with $v \in V$, is defined as a combination of three sub-sets: the	2054
vehicle types, the motor types, and the battery chemistries, resulting in 89 specific	2055
vehicles.	2056
	2057
Consider the subset of vehicle types VT , defined as:	2058
	2059
$VT = \{\text{ICEV}, \text{PHEV}, \text{BEV}\}, \quad vT \in VT.$	2060
	2061
Consider the subset of motor types V_{motor} , defined as:	2062
	2063
$V_{\text{motor}} = \{\text{PM_Cu}, \text{Ind_Cu}, \text{PM_Al}, \text{Ind_Al}\}, \quad v_{\text{motor}} \in V_{\text{motor}}.$	2064
	2065
Consider the subset of battery chemistries V_{bat} , defined as:	2066
	2067
$V_{\text{bat}} = \{\text{Gr-LMO}, \text{Gr-NMC}(111), \text{Gr-NMC}(433), \text{Gr-NMC}(532), \text{Gr-NMC}(622), \text{Gr-NMC}(811),$	2068
$\text{Gr-LMFP}, \text{Gr-NCA95}, \text{Gr-NCA85}, \text{Gr-LFP}, \text{Na-ion}\}, \quad v_{\text{bat}} \in V_{\text{bat}}.$	2069
	2070

2071 Similarly than for the energy generation technologies, a set of aggregated group
2072 of vehicles was created. The subset of vehicles V_{agg} , with $v_{\text{agg}} \in V_{\text{agg}}$, is defined as a
2073 combination of three sub-sets: the vehicle types, the aggregated motor types, and the
2074 aggregated battery chemistries, resulting in 25 combinations of aggregated vehicles.

2075
2076 Consider the subset of aggregated motor types $V_{\text{motor agg}}$,

$$2077 \quad V_{\text{motor agg}} = \{\text{PM, Ind}\}, \quad v_{\text{motor agg}} \in V_{\text{motor agg}}.$$

2078
2079 Consider the subset of aggregated battery chemistries $V_{\text{bat agg}}$, defined as:

$$2080 \quad V_{\text{bat agg}} = \{\text{Gr-LMO, Gr-NMC, Gr-LMFP, Gr-NCA, Gr-LFP, Na-ion}\}, \quad v_{\text{bat agg}} \in V_{\text{bat agg}}.$$

2084 3.6.2 Variables

2085 *The variables are defined in the optimization code in the function: modelDef.*

2086
2087 All variables are non-negative real numbers initialized at zero.

2090 **Table 44** Model variables

2092 Variable	2092 Variable description
2093 $S(r, t, d)$	2093 Cumulated installed power capacities by region r , by power 2094 plant t , at the decade d
2095 $X(v, y)$	2095 Vehicle v sold by year y
2096 $X_{\text{growth}}(vT, y)$	2096 Vehicle types vT sold to meet stock growth, at the year y
2097 $X_{\text{maintain}}(vT, y)$	2097 Vehicle types vT sold to replace the retirement of old stock 2098 at the year y
2099 $N(m, y)$	2099 Demand for electric grid by metal m and year y
2100 $\text{Res}_{\text{relax}}(m)$	2100 Relaxation variable. Simulates the creation of additional 2101 resource for the metal m , when the optimization is unable 2102 to avoid consuming all resources
2103 $\text{Mining}_{\text{relax}}(m, d)$	2103 Relaxation variable. Simulates the creation of additional 2104 mining capacity for the metal m , at the decade d , when 2105 the optimization is unable to avoid consuming all mining 2106 capacity 2107

3.6.3 Parameters

The parameters are known constant values, which are calculated before the optimization process. The calculations can be found within the code or in the Supplementary Data files.

Table 45 Model parameters

Parameter	Definition
$S_0(r, t_{\text{agg}}, d)$	Initial mix of cumulated installed power capacities by region r , by aggregated power plant t_{agg} , at the decade d
$X_0(v_{\text{agg}}, y)$	Initial mix of aggregated vehicles v_{agg} sold by year y
$X_0^{\text{growth}}(v_T, y)$	Initial vehicles sold to meet stock growth by vehicle type v_T , at the year y
$N_0(m, y)$	Initial demand for electric grid by metal m and year y
$H(t, t_{\text{agg}})$	Binary technology mapping matrix between the interchangeable sub-technologies t and the aggregated group of power plants t_{agg}
$I(v, v_{\text{agg}})$	Binary technology mapping matrix between the interchangeable sub-technologies v and the aggregated group of vehicles v_{agg}
$CF(r, t, d)$	Capacity factor of the power plant
$MI_t(m, t)$	Metal intensity of power plants
$MI_v(m, v)$	Metal intensity of vehicles
$R(m, y)$	Recycling rate of EV metals
$\text{BiomPotential}(d)$	Maximum potential of biomass power installation at the decade d
$\text{HydroPotential}(r)$	Maximum potential of hydroelectric power installation in the region r
$ST(m, y)$	Metal demand from the storage sector
$RoE(m, y)$	Metal demand from the rest of the economy
$Res(m)$	Resources of the metal m
$Prod(m, y)$	Mining capacity of the metal m at the year y
$RR_{Res}(m)$	Recovery rate of losses during production, manufacturing, and processing
$RR_{Prod}(m)$	Recovery rate of losses during manufacturing and processing

3.6.4 Objective function

Function in the optimization code: modelObj.

The objective function aims to minimize the relative squared deviations between the variables and the initial scenario. ε is set at 1 and aims to avoid divisions by 0.

For power capacities and electric vehicles, the variation between some sub-technologies is not penalized. It is done by multiplying the variable with a binary matrix corresponding interchangeable sub-technologies with an aggregated group of technology, as presented in Tab. 43.

2163 Since the model is not able to avoid all metal constraints, a relaxation variable is
 2164 set for each metal constraint (resources and mining capacity). We penalize the relative
 2165 excess to the constraint by multiplying it by an important penalization factor M , set
 2166 at 10^{15} by an iterative process.

2167 In order to avoid scaling issues while generating the code, the variable N was
 2168 considered in million of tons of metal, and multiplied by a factor of 10^6 in the objective
 2169 function. It was not included in this equation for readability purposes.

$$\begin{aligned}
 2170 \quad & \min_S \left[\sum_{r, t_{agg}, d} \left(\left(\frac{\sum_t [H(t, t_{agg}) \cdot S(r, t, d)] - S_0(r, t_{agg}, d)}{S_0(r, t_{agg}, d) + \varepsilon} \right)^2 \right) \right. \\
 2171 \quad & + \sum_{v_{agg}, y} \left(\left(\frac{\sum_v [I(v, v_{agg}) \cdot X(v, y)] - X_0(v_{agg}, y)}{X_0(v_{agg}, y) + \varepsilon} \right)^2 \right) \\
 2172 \quad & + \sum_{m, y} \left(\left(\frac{N(m, y) - N_0(m, y)}{N_0(m, y) + \varepsilon} \right)^2 \right) \\
 2173 \quad & + \sum_m \frac{M \cdot \text{Res}_{\text{relax}}(m)}{\text{Res}(m)} \\
 2174 \quad & \left. + \sum_{m, d} \frac{M \cdot \text{Mining}_{\text{relax}}(m, d)}{\text{Prod}(m, d)} \right] \tag{1}
 \end{aligned}$$

2186 3.6.5 Constraint - Energy demand

2187 *Function in the optimization code: CstrEnergyDemand.*

2188 The constraint on energy demand aims for the installed power capacity variable
 2189 $S(r, t, d)$ to meet the energy demand estimated by the IAM in the initial scenario.

$$\begin{aligned}
 2190 \quad & \sum_{t_{agg}} \left[\sum_t H(t, t_{agg}) \times S(r, t, d) \right] \times CF(r, t_{agg}, d) \\
 2191 \quad & \geq \sum_{t_{agg}} [S_0(r, t_{agg}, d) \times CF(r, t_{agg}, d)], \quad \forall r, d \tag{2}
 \end{aligned}$$

2199 3.6.6 Constraint - Capped fossil technologies

2200 *Function in the optimization code: CstrCappedCC*

2201 The constraint on capped climate change aims for the installed power capacity
 2202 variable to keep the initially installed power capacities for fossil fuel technologies.

$$2203 \quad S(r, t_{foss}, d) - S_0(r, t_{foss}, d) = 0, \quad \forall r, t_{foss}, d \tag{3}$$

3.6.7 Constraint - Biomass availability

Function in the optimization code: CstrBiomassAvail

The potential for biomass installation is greatly constrained by land availability. In order to generate a feasible power capacity mix, we constrained the variable $S(r, d, t)$ to not exceed the maximum potential of installation of biomass by decade.

$$\sum_r S(r, d, \text{Biomass}) \leq \text{BiomPotential}(d), \quad \forall d \quad (4)$$

3.6.8 Constraint - Hydro availability

Function in the optimization code: CstrHydroAvail

The potential for hydroelectricity installation is greatly constrained by regional waterfowl availability. Similarly to biomass, we constrained the variable $S(r, d, t)$ to not exceed the maximum potential of installation of hydroelectricity by region.

$$S(r, 2050, \text{Hydro}) - S(r, 2020, \text{Hydro}) \leq \text{HydroPotential}(r), \quad \forall r \quad (5)$$

3.6.9 Constraint - Technological coherence in time

Function in the optimization code: CstrTechnoCoherence

This set of constraints was determined to add coherence to the evolution of the power capacity and vehicle mix in time.

First, since the 2020 power mix has already been installed, the variable cannot change the past power mix, and is fixed to stay identical to the initial scenario.

$$S(r, t, 2020) = S_0(r, t, 2020), \quad \forall r, \forall t \quad (6)$$

Similarly than for the power capacity variable, the past EV mix from 2018 to 2020 must stay identical to the initial scenario.

$$\sum_v I(v, v_{\text{agg}}) \times X(v, y) = X_0(v_{\text{agg}}, y), \quad \forall v_{\text{agg}}, \quad \forall y \in \{2018, 2019, 2020\} \quad (7)$$

Then, to ensure temporal coherence in the evolution of power capacity infrastructures, we imposed constraints on their growth trajectories.

In most cases, the initial scenario predicts an increase in low-carbon installed capacities. For these cases, once a capacity is installed, it remains in place in subsequent decades. Thus, the power capacity variable can only increase or stay constant over time.

2255 However, in a few exceptional cases, the initial scenario projects a decline in
 2256 capacity installation due to infrastructures reaching their end of life. In these cases,
 2257 the power capacity variable is allowed to decrease but must remain above the ini-
 2258 tial scenario's capacity to avoid simulating an unrealistically accelerated retirement of
 2259 assets.

2260

2261

$$2262 \quad S(r, t, d) \geq \begin{cases} S_0(r, t, d), & \text{if } S_0(r, t, d) < S_0(r, t, d - 1) \\ S(r, t, d - 1), & \text{else} \end{cases} \quad \forall r, t, d \in \{2030, 2040, 2050\}$$

2263

2264

2265

(8)

2266 3.6.10 Constraint - Vehicle demand

2267

2268 *Function in the optimization code: CstrVehicleDemand*

2269

2270 We used a stock dynamics approach to simulate future vehicle demand. Based on
 2271 an initial vehicle stock scenario, annual vehicle sales for stock growth are equal to
 2272 the initially predicted growth $X_{0 \text{ growth}}(v_T, y)$. Annual sales for stock maintenance,
 2273 to replace vehicles reaching their end of life, are calculated based on vehicles sold 12
 2274 years earlier- the assumed vehicle lifetime L_V .

2275 The vehicle decision variable $X(v, y)$ is a sum of the sales for stock growth (X_{growth})
 2276 and the sales for stock maintenance (X_{maintain}).

2277 Since each vehicle type vT are used for distinct purposes and emit variable amounts
 2278 of greenhouse gas emissions, we preserve the initial distribution of vT as defined in
 2279 the initial scenario. To ensure this, all specific vehicle v are grouped according to their
 2280 corresponding aggregated vehicle type vT by using a summation. The specific battery
 2281 and motor used in EV remain flexible and are subject to optimization.

2282

2283

$$2284 \quad X_{\text{growth}}(v_T, y) = X_{0 \text{ growth}}(v_T, y), \quad \forall v_T, \forall y \in [2006, 2050] \quad (9)$$

$$2285 \quad X_{\text{maintain}}(v_T, y) = X_{\text{growth}}(v_T, y - L_v), \quad \forall v_T, \forall y \in [2006 + L_v, 2050]$$

2286

(10)

2287

$$2288 \quad \sum_{v \in v_T} X(v, y) = X_{\text{growth}}(v_T, y) + X_{\text{maintain}}(v_T, y), \quad \forall v_T, \forall y \in [2006 + L_v, 2050]$$

2289

(11)

2290

2291 3.6.11 Constraint - Na ion technological deployment

2292

2293 *Function in the optimization code: CstrNewTechnoEV*

2294

2295 Since Na-ion battery is an emerging technology, we constrained its penetration on
 2296 the market before 2035. Therefore, until 2035, the market shares of Na-ion batteries
 2297 were constrained to not exceed projected share from the initial scenario by more than
 2298 10%. After 2035, this constraint was lifted due to the anticipated cost advantage of
 2299 Na-ion technologies.

2300

$$\sum_{\text{Na-ion}} X(v, y) \leq 1.1 \times \sum_{\text{Na-ion}} X_0(v, y), \quad \forall y \in [2020, 2035] \quad (12)$$

3.6.12 Constraint - Grid network demand and substitution

Function in the optimization code: CstrNetworkSubstitution

To meet electric grid metal requirements, the grid variable ($N(m, y)$) has to meet the grid metal demand projected in the initial scenario. Though, since aluminum is a promising substitute for copper in wires, we added the potential of substituting 2 tons of copper by 1 ton of aluminum in the grid constraint.

$$N(\text{Aluminum}, y) \times 2 + N(\text{Copper}, y) = N_0(\text{Aluminum}, y) \times 2 + N_0(\text{Copper}, y), \forall y \quad (13)$$

3.6.13 Constraint - Cumulated demand and resources

Function in the optimization code: CstrMetal_Res

This constraint aims to limit the cumulated metal demand under metal resources. For each metal m , the cumulated demand from optimization variables plus the demand from storage and the rest of the economy between 2020 and 2050 has to remain below the resource limit, $Res(m)$.

Metal losses between the extraction of metal in reserves and the final product are taken into account in the transition sectors by dividing by the recovery rate $RR_{\text{Res}}(m)$.

In order to have a less restrictive constraint, a relaxation variable $Res_{\text{relax}}(m)$ is introduced. It allows for the metal demand to exceed resources, but with such excess being greatly penalized in the objective function.

$$\sum_{r,t} \left[\frac{MI_t(m, t)}{RR_{\text{Res}}(m)} \cdot (S(r, t, 2050) - S(r, t, 2020)) \right] \quad (14)$$

$$\sum_{v,y} \left[\frac{MI_v(m, v)}{RR_{\text{Res}}(m)} \cdot [X(v, y) - X(v, y - L_v) \cdot R(m, y)] \right] \quad (15)$$

$$+ \sum_y \left[\frac{N(m, y)}{RR_{\text{Res}}(m)} \right] + \sum_y \left[\frac{ST(m, y)}{RR_{\text{Res}}(m)} \right] + \sum_y [RoE(m, y)] \quad (16)$$

$$\leq Res(m) + Res_{\text{relax}}(m) \quad \forall m \quad (17)$$

2347 3.6.14 Constraint - Annual demand and mining capacity

2348 *Function in the optimization code: CstrMetal_Mining*
2349

2350 This constraint aims to limit the annual metal demand under mining capacity.
2351 Though, the sectors of the rest of the economy are expected to make sobriety efforts in
2352 order to reserve a share of the mining capacity to the transition. Thus, metal demand
2353 can also exceed mining capacity, if the demand from the transition sectors is under an
2354 α share of the mining capacity, set at 10% here.

2355 In order to estimate the annual metal demand from the power plant variable
2356 $S(r, d, t)$, we assume that new power capacity installations are distributed linearly over
2357 each decade. The newly installed power capacities during a decade is divided by ten
2358 to approximate the annual installed power capacity. In contrast, the vehicle variable
2359 $X(v, y)$ and the grid variable $N(m, y)$ are defined on an annual basis, while the con-
2360 straint is only applied once per decade. To ensure the constraint reflects demand over
2361 the entire period, the annual metal demand from these variables is averaged over the
2362 decade.

2363 Metal losses between the mining production and the final product are taken into
2364 account in the transition sectors by dividing by the recovery rate $RR_{\text{Prod}}(m)$.

2365 In order to make the constraint less restrictive, a relaxation variable $\text{Prod}_{\text{relax}}(m)$
2366 is introduced. It allows for the metal demand to exceed resources, but with such excess
2367 being greatly penalized in the objective function.
2368

$$\begin{aligned}
& \sum_{r,t} \left[\text{MI}_t(m, t) \times \left(\frac{S(r, t, d) - S(r, t, d-1)}{10 \cdot RR_{\text{Prod}}(m)} \right) \right] \\
& + \sum_v \left[\text{MI}_v(m, v) \times \frac{1}{10} \sum_{i=1}^{10} \left(\frac{X(v, y+i) - X(v, y+i-L_v) \cdot R(m, y)}{RR_{\text{Prod}}(m)} \right) \right] \\
& + \frac{1}{10} \sum_{i=1}^{10} \left(\frac{N(m, y+i)}{RR_{\text{Prod}}(m)} \right) + \frac{ST(m, y)}{RR_{\text{Prod}}(m)} \\
& \leq \max(Prod(m, d) - RoE(m, d), \alpha \cdot Prod(m, d)), \quad \forall m, (d, y) \in \{2030, 2040, 2050\}
\end{aligned}
\tag{18}$$

3.6.15 Abbreviations for each technologies

Table 46 Abbreviation description of technologies

Abbreviation	Full Name
Sol_C-si_Silver	Crystalline Silicon Solar with Silver conductor
Sol_C-si_Copper	Crystalline Silicon Solar with Copper conductor
Sol_CdTe	Cadmium Telluride Solar
Sol_CIGS	Copper Indium Gallium Selenide Solar
Sol_a-SiGe	Amorphous Silicon Germanium Solar
Sol_CSP_parabol_Al_MgF2	Concentrated Solar Power Parabolic, Al/MgF2 coating
Sol_CSP_tower_Al_MgF2	Concentrated Solar Power Tower, Al/MgF2 coating
Sol_CSP_parabol_Ag_GaAs	Concentrated Solar Power Parabolic, Ag/GaAs coating
Sol_CSP_tower_Ag_GaAs	Concentrated Solar Power Tower, Ag/GaAs coating
Wind_DD-EESG	Wind Direct Drive Electrically Excited Synchronous Generator
Wind_GB-DFIG_SCIG	Wind Gearbox Doubly-Fed/SC Induction Generator
Wind_DD-PMSG	Wind Direct Drive Permanent Magnet Synchronous Generator
Wind_GB-PMSG	Wind Gearbox Permanent Magnet Synchronous Generator
Nuclear_PWR	Nuclear Power Reactor (Pressurized Water Reactor)
Nuclear_HTGR	Nuclear High Temperature Gas-cooled Reactor
Nuclear_FBR	Nuclear Fast Breeder Reactor
Hydro	Hydropower
Biomass	Biomass Energy
Geothermal	Geothermal Energy
Foss_Coal	Fossil Fuel: Coal
Foss_NaturalGas	Fossil Fuel: Natural Gas
Foss_Oil	Fossil Fuel: Oil
Foss_Hydrogen	Fossil Fuel-derived Hydrogen
ICEV	Internal Combustion Engine Vehicle
PHEV	Plug-in Hybrid Electric Vehicle
BEV	Battery Electric Vehicle
PM_Cu	Permanent Magnet Motor with Copper wires
Ind_Cu	Induction Motor with Copper wires
PM_Al	Permanent Magnet Motor with Aluminum wires
Ind_Al	Induction Motor with Aluminum wires
Gr-LMO	Graphite - Lithium Manganese Oxide Battery
Gr-NMC(111)	Graphite - Lithium Nickel Manganese Cobalt Oxide (111) Battery
Gr-NMC(433)	Graphite - Lithium Nickel Manganese Cobalt Oxide (433) Battery
Gr-NMC(532)	Graphite - Lithium Nickel Manganese Cobalt Oxide (532) Battery
Gr-NMC(622)	Graphite - Lithium Nickel Manganese Cobalt Oxide (622) Battery
Gr-NMC(811)	Graphite - Lithium Nickel Manganese Cobalt Oxide (811) Battery
Gr-LMFP	Graphite - Lithium Manganese Iron Phosphate Battery
Gr-NCA95	Graphite - Lithium Nickel (95%) Cobalt Aluminum Oxide Battery
Gr-NCA85	Graphite - Lithium Nickel (85%) Cobalt Aluminum Oxide Battery
Gr-LFP	Graphite - Lithium Iron Phosphate Battery
Na-ion	Sodium-ion Battery

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